

Proof. Young's Inequality. Let $a, b \geq 0$ and let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$. We prove

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

with equality exactly when $a^p = b^q$.

1. **Conjugate exponents.** From $\frac{1}{p} + \frac{1}{q} = 1$ we obtain the familiar identities

$$q = \frac{p}{p-1}, \quad p = \frac{q}{q-1}. \quad (C)$$

2. **Degenerate cases.** If $a = 0$ or $b = 0$ then both sides of the desired inequality are zero, so the claim holds (with equality). Henceforth assume

$$a > 0, \quad b > 0. \quad (D)$$

3. **Homogenisation.** Dividing the inequality by the positive product ab yields

$$1 \leq \frac{a^{p-1}}{pb} + \frac{b^{q-1}}{qa}. \quad (1)$$

Introduce

$$u := \frac{a^{p-1}}{b}, \quad v := \frac{b^{q-1}}{a}, \quad (2)$$

so that (1) becomes

$$\frac{u}{p} + \frac{v}{q} \geq 1. \quad (3)$$

4. **Weighted AM–GM.** The weighted arithmetic–geometric mean inequality states that for non-negative x, y and weights $\lambda, \mu \geq 0$ with $\lambda + \mu = 1$,

$$\lambda x + \mu y \geq x^\lambda y^\mu,$$

with equality iff $x = y$ (or a weight equals 1). Choose

$$\lambda = \frac{1}{p}, \quad \mu = \frac{1}{q} \quad (\text{so } \lambda + \mu = 1 \text{ by (C)}),$$

and set $x = u, y = v$. Then

$$\frac{u}{p} + \frac{v}{q} \geq u^{1/p} v^{1/q}. \quad (4)$$

5. **Simplification of the right-hand side.** Using the definitions (2),

$$\begin{aligned} u^{1/p} v^{1/q} &= \left(\frac{a^{p-1}}{b} \right)^{1/p} \left(\frac{b^{q-1}}{a} \right)^{1/q} \\ &= a^{\frac{p-1}{p}} b^{-\frac{1}{p}} b^{\frac{q-1}{q}} a^{-\frac{1}{q}}. \end{aligned}$$

Because $\frac{1}{p} + \frac{1}{q} = 1$ (equivalently (C)), we have

$$a^{\frac{p-1}{p} - \frac{1}{q}} = a^{1 - \frac{1}{p} - \frac{1}{q}} = a^0 = 1, \quad b^{\frac{q-1}{q} - \frac{1}{p}} = b^0 = 1,$$

hence

$$u^{1/p} v^{1/q} = 1. \tag{5}$$

6. **Conclusion of the core inequality.** Combining (4) and (5) gives

$$\frac{u}{p} + \frac{v}{q} \geq 1,$$

which is exactly (3). Restoring u, v from (2) we obtain (1).

7. **Returning to the original variables.** Multiplying (1) by the positive quantity ab (permitted by (D)) yields

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

the desired Young's inequality.

8. **Equality condition.** Equality in the weighted AM–GM step (4) occurs iff $u = v$. From (2),

$$u = v \iff \frac{a^{p-1}}{b} = \frac{b^{q-1}}{a} \iff a^p = b^q.$$

Together with the degenerate case $a = 0$ or $b = 0$ (where both sides vanish), the complete equality condition is:

- either $a = 0$ or $b = 0$;
- or $a, b > 0$ and $a^p = b^q$.

Thus the statement is proved.

□